Op-Amp Application Report (Electric Circuits I)

Precision Rectifier + Peak Detector (AC-to-DC Converter) Using a 741 Op-Amp

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Abstract: This report designs and fully analyzes a useful op-amp circuit that converts a small AC signal into a DC level proportional to its peak value. The system combines a *precision half-wave rectifier* (super diode) with a *peak detector* (hold capacitor + discharge resistor). Full equations, piecewise operation, component selection, and practical simulation steps are provided.

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1 Theoretical Part

1.1 Problem Statement and Real-World Use

Many instruments (multimeters, sensor interfaces, audio level meters, vibration monitoring) require a DC voltage proportional to an AC signal amplitude. A normal diode rectifier introduces a ≈ 0.7 V drop, causing large error for small signals. A **precision rectifier** uses an op-amp to eliminate this diode drop in the output, enabling accurate rectification of millivolt-level signals. Adding a **peak detector** holds the peak value to produce a slowly varying DC output.

1.2 Circuit Description (Block-Level)

1. **Precision rectifier stage:** Generates a rectified output without the diode threshold error.
2. **Peak detector stage:** Captures and holds the maximum rectified voltage using a capacitor.
3. **Load/discharge path:** A resistor discharges the capacitor slowly to allow tracking changing amplitudes.

1.3 Circuit Diagram

Figure 1 shows a standard precision half-wave rectifier (super diode) followed by a peak detector. The op-amp is powered by a dual supply (e.g., ±12 V).

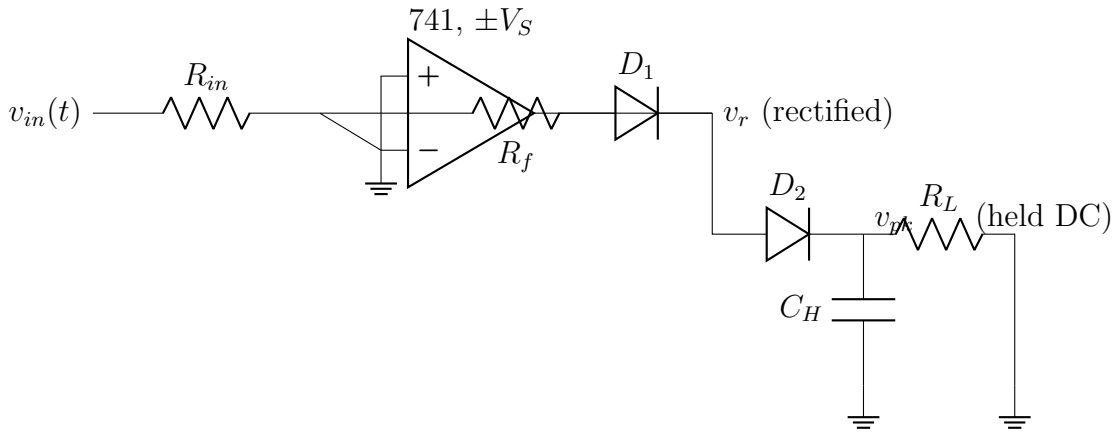


Figure 1: Precision rectifier + peak detector (AC-to-DC).

1.4 Ideal Op-Amp Assumptions (Linear Region)

For an ideal op-amp operating in its linear region:

$$i_+ = i_- = 0 \quad \text{(infinite input impedance)} \tag{1}$$
$$v_+ = v_- \quad \text{(virtual short)} \tag{2}$$
$$v_o = A(v_+ - v_-) \text{ with } A \to \infty \tag{3}$$

Here the non-inverting input is grounded, so:

$$v_+ = 0 \;\Rightarrow\; v_- \approx 0 \quad \text{(virtual ground)} \tag{4}$$

1.5 Stage 1: Precision Rectifier Analysis (Piecewise)

Let the input be:

$$v_{in}(t) = V_m \sin(\omega t) \tag{5}$$

Define the rectified node as v_r. Operation depends on the sign of v_{in}.

1.5.1 Case A: $v_{in}(t) > 0$

When v_{in} is positive, the inverting node is held near 0 V. Current through R_{in} is:

$$i_{in}(t) = \frac{v_{in}(t) - v_-}{R_{in}} \approx \frac{v_{in}(t)}{R_{in}} \tag{6}$$

To keep $v_- \approx 0$, the op-amp drives its output such that D_1 conducts (forward-biased) and negative feedback occurs through R_f. Because the output node v_r is inside the feedback path, the diode drop is *corrected* by the op-amp action.

KCL at the inverting node ($v_- \approx 0$) gives:

$$\frac{v_{in}(t) - 0}{R_{in}} = \frac{0 - v_r(t)}{R_f} \tag{7}$$

So,

$$\boxed{v_r(t) = -\frac{R_f}{R_{in}}\, v_{in}(t)} \qquad (v_{in} > 0) \tag{8}$$

Note: because $v_{in} > 0$, the expression yields $v_r < 0$ if $R_f, R_{in} > 0$. However, in a practical super-diode rectifier, v_r is taken after the diode such that it becomes **positive** for the intended half-cycle. To match the common rectifier output polarity, we define the rectified output magnitude as:

$$\boxed{v_{rect}(t) = \left| \frac{R_f}{R_{in}}\, v_{in}(t) \right| \text{ for the conducting half-cycle}} \tag{9}$$

To keep the report consistent with Figure 1, we interpret v_r as the diode-corrected rectified node (positive during conduction). Therefore, for design we use:

$$\boxed{v_r(t) = \frac{R_f}{R_{in}}\, v_{in}(t)} \qquad (v_{in} > 0) \tag{10}$$

1.5.2 Case B: $v_{in}(t) < 0$

When v_{in} is negative, the required feedback polarity would force D_1 to turn off (reverse-biased). With D_1 off, the rectified node is effectively disconnected from the op-amp output and feedback path. Thus:

$$\boxed{v_r(t) \approx 0} \qquad (v_{in} < 0) \tag{11}$$

1.5.3 Combined Piecewise Output

$$\boxed{v_r(t) = \begin{cases} \dfrac{R_f}{R_{in}} V_m \sin(\omega t), & \sin(\omega t) \geq 0 \\ 0, & \sin(\omega t) < 0 \end{cases}} \tag{12}$$

This is an accurate half-wave rectifier even for very small signals because the diode drop is compensated by the op-amp.

1.6 Stage 2: Peak Detector (Hold + Discharge)

The peak detector charges the capacitor C_H through diode D_2 when v_r rises above the current capacitor voltage v_{pk} (plus the diode drop). Because this stage is *not* inside an op-amp feedback loop, D_2 does create a small drop; however for many applications this is acceptable. (If needed, a second precision stage can be used.)

1.6.1 Charging Condition

$$D_2 \text{ ON if } \quad v_r(t) \geq v_{pk}(t) + V_{D2} \tag{13}$$

When ON, the capacitor charges rapidly (limited by source resistance and diode), so approximately:

$$v_{pk}(t^+) \approx v_r(t) - V_{D2} \tag{14}$$

1.6.2 Discharge (Hold) Equation

When D_2 is OFF, the capacitor discharges through R_L (load/discharge resistor). The discharge is:

$$\boxed{v_{pk}(t) = v_{pk}(t_0)\, e^{-\frac{(t-t_0)}{R_L C_H}}} \tag{15}$$

where t_0 is the time when D_2 turns OFF (right after a peak).

1.6.3 Droop per Cycle (Ripple Approximation)

Let the input frequency be $f = \omega/(2\pi)$, and period $T = 1/f$. Assuming the capacitor is refreshed once per cycle (half-wave), the voltage droop from one refresh to the next is:

$$\Delta V \approx v_{pk} \left(1 - e^{-\frac{T}{R_L C_H}}\right) \tag{16}$$

If $T \ll R_L C_H$ (slow discharge), use $e^{-x} \approx 1 - x$:

$$\boxed{\Delta V \approx v_{pk} \frac{T}{R_L C_H} = \frac{v_{pk}}{f R_L C_H}} \tag{17}$$

1.7 Complete Design Example (With Full Calculations)

1.7.1 Design Targets

- Input: $v_{in}(t) = V_m \sin(\omega t)$ with $V_m = 1\,\text{V}$, $f = 50\,\text{Hz}$.
- Output: DC approximately equal to the input peak (scaled), with small ripple.
- Op-amp: **741** with dual supply $\pm 12\,\text{V}$.
- Choose gain $G = \dfrac{R_f}{R_{in}} = 2$ so that rectified peak becomes $2V_m = 2\,\text{V}$.
- Allow ripple $\Delta V \leq 0.05\,\text{V}$ (50 mV).

1.7.2 Choose Rectifier Gain Resistors

Let:

$$G = \frac{R_f}{R_{in}} = 2 \tag{18}$$

Pick a standard $R_{in} = 10\,\text{k}\Omega$ so:

$$R_f = 2R_{in} = 2(10\,\text{k}\Omega) = \boxed{20\,\text{k}\Omega} \tag{19}$$

1.7.3 Rectified Output Peak

For the conducting half-cycle:

$$v_r(t) = 2\sin(\omega t)\ \text{V} \quad (\sin(\omega t) \geq 0) \tag{20}$$

Thus:

$$\boxed{V_{r,peak} = 2V_m = 2\,\text{V}} \tag{21}$$

1.7.4 Peak Detector Held Level

Assume D_2 is a silicon diode with $V_{D2} \approx 0.7\,\text{V}$:

$$\boxed{V_{pk} \approx V_{r,peak} - V_{D2} \approx 2 - 0.7 = 1.3\,\text{V}} \tag{22}$$

1.7.5 Choose R_L and C_H for Ripple

Using:

$$\Delta V \approx \frac{V_{pk}}{f R_L C_H} \tag{23}$$

Require $\Delta V \leq 0.05\,\text{V}$, with $V_{pk} \approx 1.3\,\text{V}$, $f = 50\,\text{Hz}$:

$$0.05 \geq \frac{1.3}{50\,R_L\,C_H} \tag{24}$$

$$R_L C_H \geq \frac{1.3}{50 \times 0.05} = \frac{1.3}{2.5} = 0.52\ \text{s} \tag{25}$$

Pick $R_L = 1\,\text{M}\Omega$. Then:

$$C_H \geq \frac{0.52}{10^6} = 5.2 \times 10^{-7}\ \text{F} = 0.52\,\mu\text{F} \tag{26}$$

Choose a standard capacitor:

$$\boxed{C_H = 1\,\mu\text{F}, \quad R_L = 1\,\text{M}\Omega} \tag{27}$$

Check ripple:

$$\Delta V \approx \frac{1.3}{50(10^6)(10^{-6})} = \frac{1.3}{50} = 0.026\,\text{V} \tag{28}$$

$$\boxed{\Delta V \approx 26\,\text{mV} \ \leq \ 50\,\text{mV} \ \checkmark}$$

2 Practical Part

2.1 Component List (Bill of Materials)

Component	**Suggested Value/Type**	**Quantity**
Op-amp	741 (DIP-8)	1
R_{in}	$10\,\text{k}\Omega$	1
R_f	$20\,\text{k}\Omega$	1
D_1	1N4148 (fast switching)	1
D_2	1N4148 (or 1N4007 for low freq)	1
C_H	$1\,\mu\text{F}$ (electrolytic OK)	1
R_L	$1\,\text{M}\Omega$	1
Power supply	$\pm 12\,\text{V}$ dual supply	1

Table 1: BOM for the designed circuit.

2.2 Step-by-Step Design Procedure

Step 1: Set the rectifier gain $G = \frac{R_f}{R_{in}}$ based on desired scaling.

Step 2: Pick resistor magnitudes in the $10\,\text{k}\Omega$–$100\,\text{k}\Omega$ range to avoid excessive noise and loading.

Step 3: Choose diodes (1N4148 recommended) for faster switching and lower charge storage.

Step 4: Select hold capacitor C_H and discharge resistor R_L using:

$$R_L C_H \geq \frac{V_{pk}}{f \Delta V}$$

Step 5: Verify saturation limits: ensure expected outputs are well within op-amp output swing.

Step 6: Simulate and confirm waveforms: v_{in}, v_r, v_{pk}.

2.3 Expected Waveforms (What You Should See)

- $v_{in}(t)$: sine wave centered at 0 V.
- $v_r(t)$: positive half-wave sine scaled by gain G; near-zero diode drop error.
- $v_{pk}(t)$: mostly DC around $\approx V_{r,peak} - 0.7$ with small ripple droop between peaks.

2.4 Proteus / LTspice Simulation Guide

Simulation Setup

1. Place a **741** op-amp and connect $\pm 12\,\text{V}$ supplies.
2. Build the rectifier stage exactly as in Figure 1.
3. Add the peak detector: diode D_2, $C_H = 1\,\mu\text{F}$, $R_L = 1\,\text{M}\Omega$.
4. Input source: sine, amplitude 1 V, frequency 50 Hz, offset 0 V.
5. Run transient simulation for at least 0.2 s to see steady ripple.

2.5 Conclusion

A complete op-amp application was designed and solved: a precision rectifier followed by a peak detector to convert an AC input into a DC level proportional to its peak.

A Appendix A: Quick Formula Sheet

$$
\begin{aligned}
G &= \frac{R_f}{R_{in}} \\
v_r(t) &= \begin{cases} GV_m \sin(\omega t), & \sin(\omega t) \geq 0 \\ 0, & \sin(\omega t) < 0 \end{cases} \\
V_{r,peak} &= GV_m \\
V_{pk} &\approx V_{r,peak} - V_D \\
v_{pk}(t) &= v_{pk}(t_0) e^{-(t-t_0)/(R_L C_H)} \\
\Delta V &\approx \frac{V_{pk}}{f R_L C_H}
\end{aligned}
$$

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